Fiscal Year:	FY 2006	**Task Last Updated:**	FY 06/05/2006
PI Name:	Thiessen, David Brian Ph.D.		
Project Title:	Supported Capillary Pipes For Countercurrent Two-Phase Contacting		
Division Name:	Physical Sciences		
Program/Discipline:	FLUID PHYSICS		
Program/Discipline--Element/Subdiscipline:	FLUID PHYSICS--Interfacial phenomena		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	thiessen@wsu.edu	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	509-335-4908
Organization Name:	Washington State University		
PI Address 1:	Department of Physics and Astronomy		
PI Address 2:	1405 Northeast College Ave, MS 2814		
PI Web Page:			
City:	Pullman	**State:**	WA
Zip Code:	99164-2814	**Congressional District:**	5
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2001 Physical Sciences 01-OBPR-08
Start Date:	12/01/2003	**End Date:**	11/30/2006
No. of Post Docs:	0	**No. of PhD Degrees:**	
No. of PhD Candidates:	0	**No. of Master' Degrees:**	
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA GRC
Contact Monitor:	Green, Robert D	**Contact Phone:**	216-433-5402
Contact Email:	robert.d.green@nasa.gov		
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Marston, Philip (Washington State University)		
Grant/Contract No.:	NNC04GA65G		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	It is proposed to examine the stability and dynamics of supported capillary structures in low gravity which would have application to chemical processes necessary for life support. Countercurrent flow of gas and liquid or of two liquids is ubiquitous in chemical processing. Examples include distillation, adsorption, stripping, and extraction. A large amount of surface area is necessary for efficient operation of such processes. On earth, gravity drives the countercurrent flow because of a density difference between the two phases. Various support structures are used to create large amounts of surface area between the two phases such as structured packings or bubble cap trays. In zero gravity, free-surface liquid flow can be inertial or driven by capillary forces or other applied forces. It is proposed here to examine the use of supported capillary pipes, which could sustain sufficient pressure gradients to drive flow over significant distances for liquid-gas or liquid-liquid contacting. Liquid tubes of arbitrary length can be supported by parallel wires with the proper wetting characteristics arranged in regular polygonal shapes or by a helical coil of wire. The static stability of these		

	structures has recently been investigated theoretically. Some preliminary demonstrations have been carried out in a Plateau tank. Capillary pipes might also be useful in capillary-pumped loops for heat-transfer applications. Plateau-tank experiments are planned to test stability and flow characteristics of such capillary pipes. Task objectives: 1. Determine the stability characteristics of supported capillary pipes under static conditions. 2. Study the flow characteristics in capillary pipes including initiation, steady-state flow and the effect of pressure transients. 3. Study capillary oscillations on a supported capillary pipe. 4. Explore the ability to stabilize capillary pipes using dielectrophoretic forces. The fluid configurations studied here might be useful in life support processes on long-duration manned space missions. In particular the weight of processing equipment might be reduced by using supported capillary structures.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The structures studied here might be useful for microfluidic devices which are currently of significant interest for portable medical devices and other applications on earth.
Task Progress:	Significant progress has been made toward the goal of determining the utility of supported capillary pipes for two-phase contacting in zero gravity. Progress on the specific objectives of the proposal are summarized as follows: Objective 1: Determine the stability characteristics of supported capillary pipes under static conditions. Progress: An extensive computational study was carried out to predict the maximum and minimum stable volumes and corresponding pressures in a single-wire helical support over the full range of stable helix pitches, for a range of dimensionless wire radii and for a large range of contact angles. The computed stability envelopes are strictly valid only for infinitely long liquid channels although these stability limits are expected to apply approximately to finite length liquid channels with the free surfaces pinned at the ends. Two types of instability are theoretically predicted for infinite channels depending on the wire radius, pitch and contact angle. One is a pressure turning point instability. This type of instability can occur at both a high volume limit and a low volume limit. At the upper critical volume, a slight local increase in the volume per unit length of the column leads to a locally smaller pressure thus driving a flow that causes the bulge to grow. Conversely, at the lower critical volume a slight local decrease in the volume per unit length leads to a higher local pressure driving flow away from the necked region and leading to breakage of the column. A second type of instability involves contact line merging on the outside or inside of the wire. In certain cases, when the volume per length of liquid in the support structure is large, the contact lines are predicted to meet on the outside of the wire. Experimentally this is observed to happen locally leading to local wire engulfment and a blowout of the surface. Contact line merging at low volumes is also predicted in certain cases which would be expected to lead to liquid detachment from the wire and subsequent column breakage. For contact angles near 90 degrees, the high- and low-volume instabilities are of the pressure turning point variety for the wire radii considered. For contact angles somewhat less than 90 (wetting case) the upper stability limit is reduced by the wire engulfment instability while for contact angles somewhat greater than 90 (nonwetting case) the lower volume limit is dictated by wire detachment. When the ends of a liquid segment in a support structure are free to move, the pressure in the segment is fixed. In this case the volume per unit length is not constrained but the segment length adjusts itself to minimize free energy. A theory was developed to predict this pressure and the corresponding volume per unit length for a free-ended liquid segment in a helical support. Thus the theory predicts three important pressures associated with a particular support geometry and contact angle: the maximum and minimum stable pressures of an infinitely long column and the pressure of a free-ended segment. The significance of these pressures is that they are related to the driving force available to drive a flow in the structure. The difference between the maximum and minimum stable pressure is expected to give the maximum possible driving force for flow in a support structure of a given length. The difference between the maximum pressure and free-ended segment pressure gives the driving force for flow during initial penetration of the meniscus into the support. Free-ended segments of liquid have been predicted to be stable over a more limited range of helix pitch than infinite liquid columns. This means that for pitches above a critical value it is not possible to fill the structure by injection from one end. A stable liquid column beyond this critical pitch can be formed by stretching a spring already filled with liquid. A Plateau-tank facility has been constructed to study the stability of supported capillary pipes with length to diameter ratios up to 56 for 0.25 inch diameter springs. The upper and lower stable volumes have been measured as a function of helix pitch for a particular fluid system and spring. To date, the measured stability limits are in qualitative agreement with predictions but generally show a reduced range of stability relative to predictions. This may be caused by contact angle hysteresis, contact angle nonuniformity along the wire or nonuniformity in the pitch of the helix. The equilibrium configuration of a free-ended liquid segment can also be measured. The prediction of a critical pitch beyond which the structure cannot be filled by injection from one end has also been experimentally confirmed. It has been found that the point of blowout or breakage occurs at a reproducible location for a given spring most likely because of some imperfection in the pitch or wetting. This allows for capturing the failure with high-speed video. The high-speed video records confirm that there are two failure mechanisms as predicted, one in which the initial failure occurs by wire engulfment or detachment and the other in which a local bulge or neck grows catastrophically. Objective 2: Study the flow characteristics in capillary pipes including initiation, steady-state flow and the effect of pressure transients. Progress: The flow of pure water from a reservoir into a 9 inch long, 1/8 inch diameter horizontal spring surrounded by air has been demonstrated. Such flows are possible for horizontal springs of sufficiently small diameter and pitch. Both steady flow and initial penetration flow have been studied. The rate of initial penetration of the meniscus into the structure was measured for a constant entrance pressure giving a penetration rate of around 1 cm/s falling off slightly by the time the meniscus had penetrated 9 inches. After the liquid channel had been established a steady flow of 10 mL/min could be maintained indefinitely. A continuous flow loop has also been demonstrated using a single pump to simultaneously suck liquid from one end of the structure and reinject it at the other end. Flow rate limits have not yet been measured. Objective 3: Study capillary oscillations on a supported capillary pipe. Progress: No detailed study of this has yet been made. Objective 4: Explore the ability to stabilize capillary pipes using dielectrophoretic forces. Progress: This has not yet been attempted. Progress toward practical applications: One of the envisioned applications of fluid flow in helical support structures is for gas-liquid contacting, for instance for chemical processing or heat transfer in zero gravity. To demonstrate the potential for such applications an experiment was performed to absorb CO2 with 1 M NaOH flowing in a horizontal 1/8 inch diameter, 11 inch long spring with tight

	pitch. The spring was enclosed in a plexiglas tube with sealed ends containing inlet and outlet ports for liquid and gas. The gas phase consisted of pure CO2 and a bubble meter was used to measure the rate of uptake of CO2 as a function of time. The steady liquid flow rate in the structure was 1.77 mL/min and the rate of gas absorption was 8.4 mL/min. The flow loop contained a 100 mL reservoir such that the absorption rate dropped off only slightly over a period of 1000 seconds.
Bibliography Type:	Description: (Last Updated: 09/14/2018)
Abstracts for Journals and Proceedings	Bernate J, Thiessen DB, Marston PL. "The Static Stability of Supported Capillary Pipes at Zero Bond Number." APS/DFD 58th Annual Meeting, Chicago, IL, Nov. 20-22, 2005. Bulletin of the American Physical Society DFD05, November 2005. , Nov-2005
Articles in Peer-reviewed Journals	Lowry BJ, Thiessen DB. "Fixed contact line helical interfaces in zero gravity." Phys Fluids (in press), 2006. , Jan-2006
Papers from Meeting Proceedings	Thiessen DB, Bernate J, Marston, PL. "Capillary-driven flow in supported capillary pipes." AIChE 2005 Annual Meeting Conference Proceedings, Cincinnati, OH, Oct. 30-Nov. 4, 2005. AIChE 2005 Annual Meeting Conference Proceedings, paper 124f, November 2005. , Nov-2005